EXP 1

Code:

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>

int main() {
    char text[200];
    int k, i;

    printf("Enter plaintext: ");
    fgets(text, sizeof(text), stdin);

    printf("Enter key (1-25): ");
    scanf("%d", &k);

    if (k < 1 || k > 25) {
        printf("Invalid key!\n");
        return 1;
    }

    for (i = 0; text[i] != '\0'; i++) {
        if (isupper((unsigned char)text[i]))
            text[i] = (text[i] - 'A' + k) % 26 + 'A';
        else if (islower((unsigned char)text[i]))
            text[i] = (text[i] - 'a' + k) % 26 + 'a';
    }
```

```
    printf("Ciphertext: %s", text);
    return 0;
}
```

Out Put:

```
Enter plaintext: REDD
Enter key (1-25): 4
Ciphertext: VIHH
```